

Problem 4 (25 points). Here are all the positive multiples of 9 smaller than 100 written out in Lutos, in random order:

- | | |
|--|------------------------------|
| a. b'u zo ^m bakad'ε | g. unda so zo b'u |
| b. kal b'u | h. unda so zo b'u zo kal b'u |
| c. unda d'oi zo b'u zo mi zo d'oi | i. unda so zo d'oi |
| d. unda d'oi zo kal ^m bakad'ε | j. unda zio zo b'u zo so |
| e. unda muta zo b'u zo zio | k. unda zio zo mi |
| f. unda muta zo muta | |

(a) Make the correct correspondences.

(b) Write the following equations with numbers:

(1) $\text{^mbakad'ε} \times \text{unda d'oi zo mi} = \text{^mbambo zio}$

(2) $\text{unda so zo d'oi} + \text{unda muta zo b'u zo kal b'u} = \text{^mbambo d'oi zo unda muta}$

(c) Write in Lutos: 16, 20, 39, 46, 100, 363, 477, 800.

△ Lutos (or Ruto) belongs to the Bongo-Bagirmi branch of the Central Sudanic languages. It is spoken by approx. 19,000 people in the Central African Republic and Chad.

b', **d'**, and ^m**b** are consonants; **ε** is a vowel.

The language has two tones, but they were not considered in this problem for the sake of simplicity.

—Vlad A. Neacșu

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Good luck!